Rishi Bharadwaj

Rishi-Bharadwaj | Google Scholar | rishib62107@gmail.com | +91 7975500661

Education

B.E. (Hons.) in Computer Science BITS Pilani, Hyderabad Campus **CGPA: 9.33/10** 2022–2026
Minor in Finance

Research Interests

Game Theory and Mechanism Design, Multi-Agent Systems, Reinforcement Learning, and AI for Good.

Research

Research Associate, Game Theory Lab, IISc Bengaluru Aug 2026 – Present

Working with Prof. Y. Narahari at the intersection of Game Theory, Mechanism Design, Reinforcement Learning, and social impact issues.

- **Carbon farming contracts**: Modeled carbon farming as a POMDP in which a Reinforcement Learning principal offers a single contract to a heterogeneous, partially observed population under multi-season dynamics. On large farms the aggregator realized 87.7% of achievable adoption, against only 8.2% on smallholdings. Our results and open-source simulator can guide contract and policy design that opens carbon income to smallholders while enabling agricultural soils to contribute to climate mitigation at scale. Beyond climate finance, this work contributes directly to the growing subfield of automated contract design [P1].
- **Kant-Q**: Developing a Q-learning algorithm for Kantian equilibrium, a solution concept that differs fundamentally from existing Nash-based approaches. It provides a promising foundation for cooperative AI, allowing agents to reach mutually beneficial outcomes in many social dilemmas.
- **Agricultural Yield Optimization**: Formulated a novel methodology for improved agricultural monitoring and yield optimization. Built a working model and drafted an invention disclosure. Currently under legal review.

Undergraduate Researcher, BITS Pilani, Hyderabad Campus 2024 – 2026

Advisor: Prof. Manik Gupta

- **Air Quality Arena (AQA)**: Built, to our knowledge, the largest and most geographically diverse ML-ready air quality dataset (14,139 series, 6 pollutants, 7 countries). Benchmarked 11 leading time-series foundation models (TSFMs) and classical baselines on it. Showed that zero-shot TSFMs generally outperform baselines trained directly on the target series. Accuracy was weakest in India and Mexico, making AQA a more representative evaluation framework than Europe- and US-only benchmarks [W1].
- **Air pollution and urbanization**: Analyzed five years of satellite data (Sentinel-5P, VIIRS nighttime lights, MODIS land surface temperature, NDVI) to relate air pollution to urbanization in peri-urban Hyderabad [C1].

Research Intern, Game Theory Lab, IISc Bengaluru May 2025 – Jul 2025

- Conceived and built, end to end, a multi-agent portfolio management system under Prof. Y. Narahari, in which specialized LLM analyst agents (news, fundamental, technical) propose trades to a reinforcement learning coordinator that makes portfolio allocation decisions.
- Measured the value of the LLM analysis by comparing against the same RL agent trained only on raw market indicators. Over a 6-month backtest, the multi-agent system earned a 2.3% higher cumulative return.

Publications and Preprints

[C1] Ravi Bhushan, **Rishi Bharadwaj**, Manik Gupta, Lavanya Suresh, and Timothy Clune. "Tracking Air Pollution in Peri-Urban Hyderabad, India Using Remote Sensing Data". In: *2025 IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*. 2025, pp. 1012–1016. Paper | Code.

[W1] **Rishi Bharadwaj**, Manik Gupta, and Pandarasamy Arjunan. "Air Quality Arena: A Large-Scale Multi-Region Ground Monitoring Dataset and Benchmark for Air Quality Forecasting with Time-Series Foundation Models". In: *2nd ICML Workshop on Foundation Models for Structured Data*. Non-archival. 2026. Paper | Code | Dataset.

[P1] **Rishi Bharadwaj** and Yadati Narahari. *Learning Principal-Agent Contracts for Equitable Smallholder Carbon Farming under Moral Hazard and Adverse Selection*. 2026. arXiv: 2609.20404 [cs.LG]. Under review. Paper | Code.

Presentations

Poster, 2nd ICML Workshop on Foundation Models for Structured Data [W1] 2026

Industry Experience

Applied AI/ML Intern, JPMorgan Chase & Co. Jul 2025 – Dec 2025

- Built an end-to-end LLM pipeline to reduce manual handling of client inquiries on settled payments. In evaluation, it resolved 80% of in-scope inquiries without manual review (straight-through processing) at >90% accuracy.
- Included intent recognition and entity extraction for unstructured client messages in the pipeline, with consistency checks that cross-verify extracted fields against source data to mitigate hallucinations.

Machine Learning Intern, Voicegain.ai May 2024 – Jul 2024

- Benchmarked open-source embedding models and selected bge-small, the best performer (Spearman 0.83 on STS13 vs 0.50 for the in-house encoder). Helped deploy it internally on NVIDIA Triton Inference Server, where it ran 3× faster than the model it replaced.
- Reused the model to prototype intent recognition for call-center transcripts as a cheaper alternative to OpenAI embeddings. Matched customer names across speech-to-text spelling errors with Soundex and Levenshtein distance.

Selected Coursework

Mathematics I (Calculus), Mathematics II (Linear Algebra), Probability and Statistics, Data Structures and Algorithms, Design and Analysis of Algorithms, Machine Learning, Principles of Economics

Extracurriculars

Lead, Help Desk, Department of Controlz, BITS Pilani, Hyderabad Campus 2024–2026

- Led a 6-person team running the festival help desk across four festivals.

Animal Care Volunteer, BITS Pilani, Hyderabad Campus 2023–2026

- Active member of the campus animal care community. Helped care for campus cats, including accompanying them on veterinary visits and helping organize funding for their care.

Awards and Honors

Merit Scholarship, BITS Pilani (awarded every semester, 50–80% of tuition) 2022–2026
State Rank 9, Karnataka Class XII Board Examinations (98.3%) 2022